## SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

### Department of Computer Science and Engineering

### ASSESSMENT TOOL 2– Scenario-Based

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|---|---|---|---|
| **Course Code:** | ITA0512 | **Course Title:** | Computer Vision |
| **CO Assessed:** | CO2 – Imitation (BL3) | **Assessment:** | Assessment Tool 1 – **Scenario-Based** |
| **Weightage:** | 10% of CIA | **Total Marks:** | 100 |
| **CO2 Statement:** | *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)* | | |
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| **Batch** | B. Tech AI & ML | **Date:** | |

**Code of Conduct**

*I, S. Susannal Devapriyam (192525398) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: *S. Susannal Devapriyam*

### RUBRICS FOR CALCULATION: ASSESSMENT TOOL 2: Scenario-Based

| Criteria | Weight age | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning | Marks |
|---|---|---|---|---|---|---|
| **Understanding of Scenario** | **20%** | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps | Partial understanding; misses key aspects | Misinterprets or fails to understand the scenario | |
| **Identification of Key Issues** | **20%** | Accurately identifies all relevant issues and factors involved | Identifies most key issues with minor omissions | Identifies limited issues; some irrelevant points | Unable to identify key issues | |
| **Application of Concepts** | **25%** | Applies appropriate concepts effectively to address the scenario | Applies concepts with minor errors | Limited or partially correct application | Unable to apply relevant concepts | |
| **Decision Making** | **20%** | Makes well-reasoned, logical decisions with strong rationale | Makes reasonable decisions with some justification | Makes weak decisions with limited justification | No clear decisions made or rationale provided | |
| **Clarity of Response** | **15%** | Response is clear, structured, and logically presented | Generally clear with minor issues | Somewhat unclear or poorly structured | Disorganized and difficult to understand | |

## Question :

## 9. Broken Edges

**After applying edge detection, object boundaries appear discontinuous.**

**(a)** Interpret the scenario and explain why edges are broken.

**(b)** Identify key issues affecting boundary continuity.

**(c)** Apply a suitable method to improve edge connectivity.

**(d)** Justify your approach logically.

**(e)** Present your explanation clearly.

## Answer :

### (a) Interpret the Scenario and Explain Why Edges are Broken

#### Problem Understanding

Edge detection is an important process in Computer Vision used to identify object boundaries by detecting sudden changes in pixel intensity. Ideally, the detected edges should form continuous boundaries around objects. However, in this scenario, the detected edges appear broken or discontinuous.

Broken edges occur when some edge pixels are not detected during the edge detection process. Instead of obtaining a complete boundary, only small disconnected edge segments are produced. This makes it difficult to recognize the exact shape and size of the object.

#### Reasons for Broken Edges

- Presence of image noise causes false or missing edge pixels.
- Low image contrast reduces the intensity difference between object and background.
- Improper threshold selection removes weak but important edge pixels.
- Sensor imperfections during image acquisition introduce distortions.

- Poor lighting conditions reduce edge visibility.
- Compression artifacts or image blur weaken object boundaries.

### (b) Identify Key Issues Affecting Boundary Continuity

Several factors reduce the continuity of object boundaries.

#### 1. Image Noise

Random variations in pixel intensity interrupt continuous edges and produce unwanted edge fragments.

#### 2. Low Contrast

When neighbouring pixels have very similar intensity values, the edge detector cannot distinguish object boundaries clearly.

#### 3. Improper Threshold Value

A very high threshold removes many valid edges, whereas a very low threshold introduces unwanted noisy edges.

#### 4. Weak Edge Magnitude

Gradual intensity changes generate weak edges that are often discarded during thresholding.

#### 5. Image Blur

Motion blur or defocus blur smoothens intensity transitions, making edges less detectable.

#### 6. Inaccurate Edge Detection Operator

Some operators may fail to detect weak edges depending on image characteristics.

### (c) Apply a Suitable Method to Improve Edge Connectivity

A suitable method for improving edge connectivity is the **Morphological Closing Operation**.

Morphological Closing is performed using two operations:

1. **Dilation**
2. **Erosion**

### Step 1 – Original Broken Edge

```
########  #######
    ##      ##
```

The object boundary contains visible gaps.

### Step 2 – Dilation

Dilation expands edge pixels.

```
###############
###############
```

Small gaps between neighbouring edge pixels become connected.

### Step 3 – Erosion

Erosion removes extra pixels added during dilation while preserving the newly connected boundary.

```
###############
```

The final object boundary becomes continuous and complete.

### Alternative Method

Another effective method is **Canny Edge Detection**.

The Canny algorithm uses:

- Gaussian filtering for noise removal.
- Gradient calculation.
- Non-Maximum Suppression.

- Double Thresholding.
- Edge Tracking by Hysteresis.

These stages preserve weak edges connected to strong edges and produce continuous object boundaries.

**(d) Justify the Approach Logically**

Morphological Closing is selected because it directly fills small gaps between disconnected edge pixels without significantly changing the overall object shape.

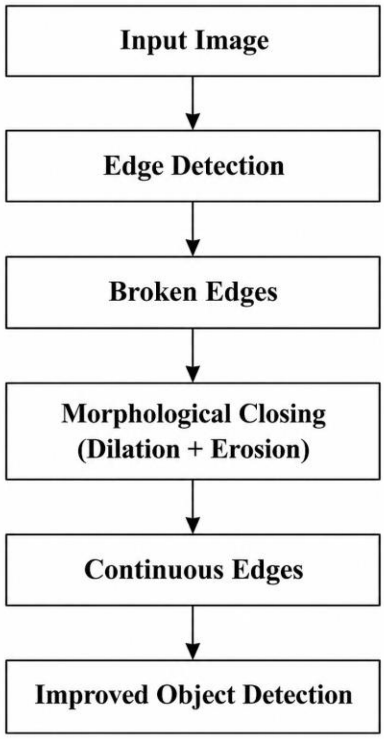
Its advantages include:

- Improves edge continuity.
- Connects neighbouring edge segments.
- Preserves object boundaries.
- Removes small discontinuities.
- Produces smoother segmentation results.
- Enhances feature extraction.
- Improves object recognition accuracy.

If the Canny algorithm is used, edge tracking by hysteresis ensures that weak edges connected to strong edges are retained, thereby reducing broken boundaries.

## (e) Present the Explanation Clearly

Flow of Improvement




### Result

The discontinuous boundaries become continuous after applying Morphological Closing. Connected edges improve object segmentation, feature extraction, shape analysis, and object recognition. Continuous boundaries also reduce segmentation errors and improve the overall performance of Computer Vision systems.

### Conclusion

Broken edges are mainly caused by noise, low contrast, improper threshold selection, and image blur. Morphological Closing effectively connects discontinuous edge segments by combining dilation and erosion operations. As a result, object boundaries become continuous, leading to more accurate segmentation, feature extraction, and object recognition. Therefore, Morphological Closing is a simple and effective technique for improving edge connectivity in Computer Vision applications.